EF

EXTREME FITNESS QUALITY ASSURANCE DOCUMENT

BIT507 Web Applications Implementation

Web Development Project: Responsive Interactive Website

Student name	[Your full name]
Student ID	[Your student ID]
Website	Extreme Fitness Gym (fictional), a five-page static website: `index.html`, `classes.html`, `membership.html`, `join.html`, `thank-you.html`
Technologies	HTML, CSS, vanilla JavaScript (no frameworks or build tools); Archivo from Google Fonts
Date	October 2026
Word count	2,695 words (sections 2 to 9, excluding tables, figure captions, references and appendix)

1. TABLE OF CONTENTS

2. Conceptualisation **3**

3. Design **4**

3a. Site map 4

3b. Wireframes 4

3c. Colour scheme 6

4. Content **8**

4a. Home (index.html) 8

4b. Classes (classes.html) 9

4c. Membership (membership.html) 10

4d. Book a free trial (join.html) 11

5. Web Design Best Practices **12**

5a. Responsive page layout 12

5b. Navigation 12

5c. Accessibility 13

6. Testing **15**

6a. Browser a: Google Chrome 15

6b. Browser b: Microsoft Edge 16

6c. Screen size a: 320px (small phone) 16

6d. Screen size b: 768px (tablet portrait) 17

6e. Screen size c: 1366px (laptop) 18

6f. Mobile Lighthouse audit 18

6g. HTML/CSS validator 19

6h. Bugs found and fixed 20

7. Launch **22**

8. Maintenance **23**

8a. Accessibility 23

8b. Feature improvements 23

9. Conclusion **24**

10. Referencing **25**

Appendix A: AI Declaration **26**

2. CONCEPTUALISATION

Extreme Fitness is a five-page static website for a fictional, beginner-friendly gym. The idea came from noticing how intimidating big commercial gyms can be. My target audience is adults aged 18 to 45 who are new to lifting or getting back into it after a long break. They are usually browsing on their phones, trying to work out whether they actually want to walk through the doors.

The main problem I wanted to solve is that small gyms often hide their class timetables and pricing in messy PDFs or Instagram posts. Beginners really want to know three things. Which classes will not make them look foolish? What does it cost? What actually happens on day one? A clean, mobile-first site answers those questions quickly.

Motivation

I have been going to the gym for a few years and still remember how overwhelming it was at first. I wanted to build something that felt welcoming rather than aggressive.

Objectives

- Let a first-timer filter the timetable to find beginner-friendly sessions.
- Break down membership costs and answer common worries.
- Provide a working, validated form to request a free trial.
- Make sure the layout does not break or force sideways scrolling on phones, tablets or desktops.
- Keep the site usable with only a keyboard, and make sure the core content still makes sense if CSS or JavaScript fails to load.

Scope and limitations

This is strictly a front-end project using HTML, CSS and vanilla JavaScript. There is no backend, database or payment processing. The trial form uses `method="get"` to send every filled-in field to a confirmation page. The gym, address, phone number and email are fictional. I used generative AI to help draft placeholder text and, later, to check this document against my code; both are declared in Appendix A.

3. DESIGN

The site has four main pages and a confirmation page. They all share the same header, footer, stylesheet and script, so users always know where they are. Files are organised by type with lowercase, hyphenated names and relative paths, so the site runs straight from the extracted ZIP.

```
sree_lakshmi_web/
├── index.html          Home
├── classes.html        Class timetable (filter)
├── membership.html     Plans + FAQ (accordion)
├── join.html           Book a free trial (form)
├── thank-you.html      Confirmation (form target, not in menu)
├── css/styles.css      One mobile-first stylesheet
├── js/script.js        Menu toggle, class filter, FAQ accordion
└── assets/images/      hero-barbell-squat-800.webp, -1600.webp, favicon.svg
```

3a. Site map

Home is the entry point. The header and footer link to the four main pages from every page. The trial form on the Join page submits to `thank-you.html`, which is not in the menu. It links back to the timetable, the membership FAQ and Home, so the visitor is never stuck.

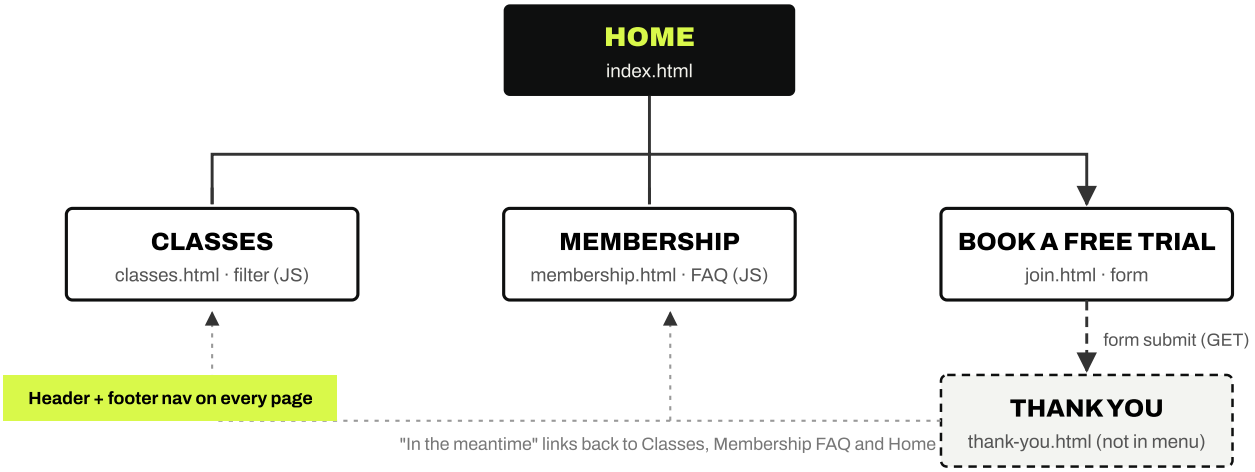


Figure 1. Site map. Solid lines are the main navigation, available in the header and footer of every page. The dashed line is the form submission to the confirmation page.

3b. Wireframes

I planned everything mobile-first. On a phone each page is a single column, and wider screens get extra columns only where the content needs them. I changed one thing during development. My original mobile wireframe for Home put the photo above the heading. In the build I swapped them so the heading and the "Book a free trial" button appear without scrolling. On the Join page I used CSS Grid template areas to move "What happens on your first visit" and "Find us" into a right-hand column on desktop without changing the HTML order. This keeps the tab and screen-reader order the same as on mobile.



Figure 2. Home wireframes. The mobile plan shows the photo first; the note marks the change made in the build.

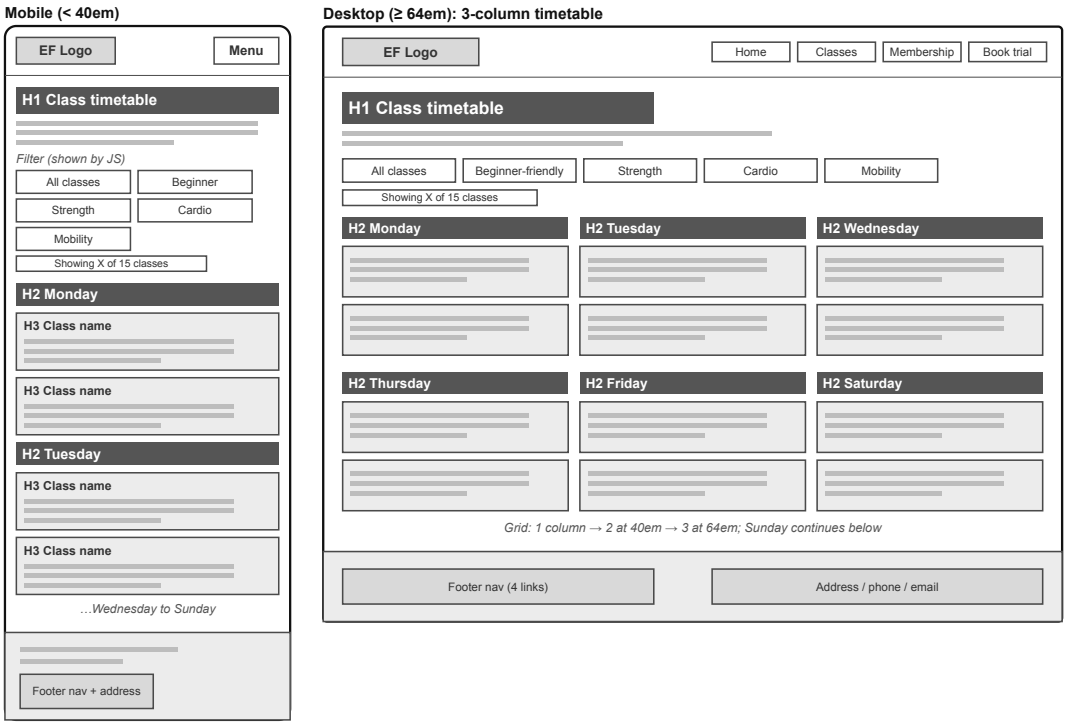


Figure 3. Classes wireframes. The filter bar sits above day sections that become a 2-column and then a 3-column grid.

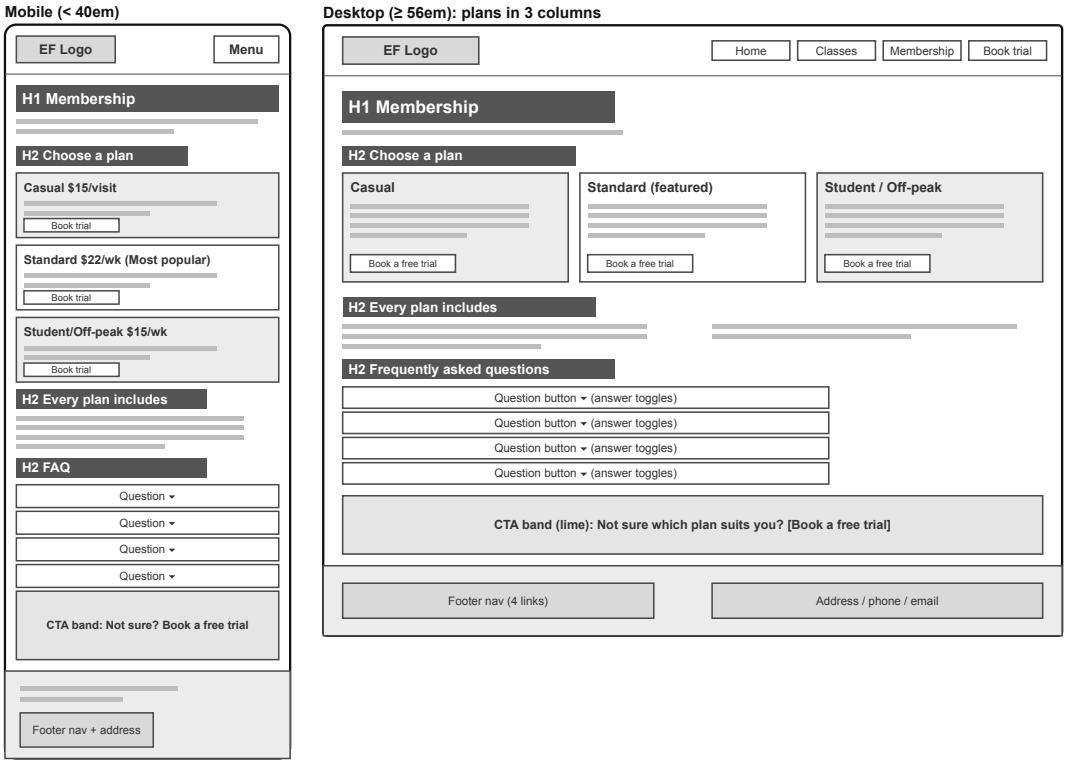


Figure 4. Membership wireframes. Plan cards, an "Every plan includes" list, the FAQ accordion and a call-to-action band.

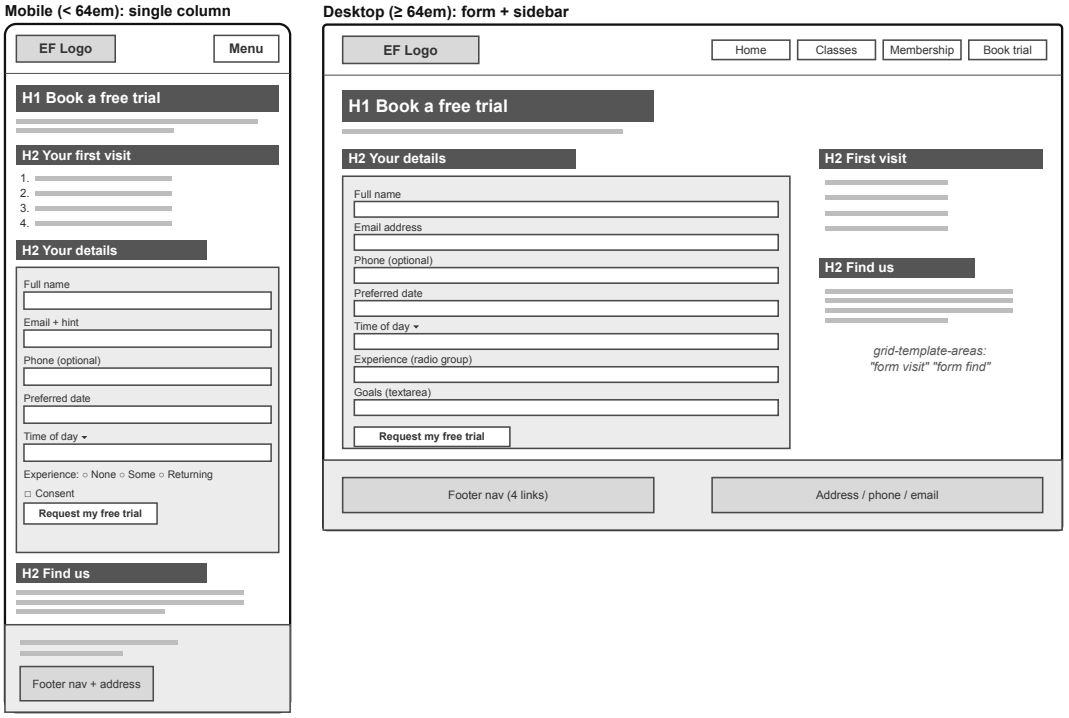


Figure 5. Join (Book a free trial) wireframes. On desktop the form spans two rows next to the first-visit steps and the "Find us" details.

3c. Colour scheme

I went with a dark theme and a single lime accent. I first looked at a reference site called Iron Peak Gym, which uses a dark layout with bold uppercase type and an orange accent. I liked the high-contrast feel but swapped orange for lime to give my brand its own identity. My first palette used light charcoal, teal and amber. When I tested the amber text on the light background it only reached 2.00:1, so I dropped it. All final colours are CSS custom properties in `:root`, and I ran every pair through the WCAG 2 contrast formula (Table 1). The lowest text pair is the muted grey on the card surface at 7.62:1, well above the 4.5:1 minimum. Control borders reach 3.73:1, above the 3:1 needed for interface components.

Table 1. Final colour palette and measured contrast ratios (WCAG 2 relative-luminance formula)

Colour	Token	Hex	Used for	Contrast
	`--color-bg`	#0E0F0F	Page background; text on lime	Base
	`--color-surface`	#181919	Cards, form panel, footer, hours table	Base
	`--color-surface-2`	#202121	Inputs, class level tags	Base
	`--color-line`	#2E2F2F	Dividers and card borders (decorative)	n/a
	`--color-text`	#F2F2EC	Body text, headings	17.08:1 on bg · 15.68:1 on surface · 14.36:1 on surface-2
	`--color-muted`	#ABABA3	Secondary text, hints, captions	8.31:1 on bg · 7.62:1 on surface
	`--color-accent`	#D9F94A	Links, buttons, active states, focus ring	16.09:1 on bg · 13.53:1 on surface-2
	`--color-on-accent`	#0E0F0F	Text on lime buttons and the CTA band	16.09:1 on accent
	`--color-input-border`	#7A7A73	Input, filter and Menu button borders	3.73:1 on surface-2 (non-text, needs 3:1)

Typography

I used Archivo from Google Fonts, loaded as a variable font with a width axis (62 to 125%) and a weight axis (400 to 900). Headings are weight 900, condensed to 75% width and uppercase so they feel athletic. Body text is normal width, 17px (1.0625rem) with a 1.6 line height, and paragraphs are capped at 65ch so lines stay easy to read. Heading sizes use `clamp()` so they scale smoothly with the screen.

Responsive approach

The CSS is mobile-first. The base styles are the phone layout, and four `min-width` media queries in em units add columns where the content started to look cramped or stretched (Table 2). Grid handles the page sections, cards and timetable; Flexbox handles the header, navigation and filter buttons.

Table 2. Content-led breakpoints in css/styles.css

Breakpoint	≈ px	What changes
Base	< 640	Single column; "Menu" button reveals the navigation; filter buttons wrap; hero photo 4:3 under the heading
`40em`	640	Timetable 2 columns; "Every plan includes" list in 2 columns; footer in 2 columns; more padding in the CTA band
`52em`	832	Menu button hidden and the full navigation shown in the header (moved from 48em, see 6h)
`56em`	896	Hero text beside a square photo; Home cards and membership plans in 3 columns; hours and next-step side by side
`64em`	1024	Timetable 3 columns; Join page form-plus-sidebar grid; extra vertical padding; FAQ limited to 50rem wide

4. CONTENT

Each page tackles one of a beginner's main questions and ends with a route to the free trial. The screenshots are full-page captures in Google Chrome at a 1366px-wide window.

4a. Home (index.html)

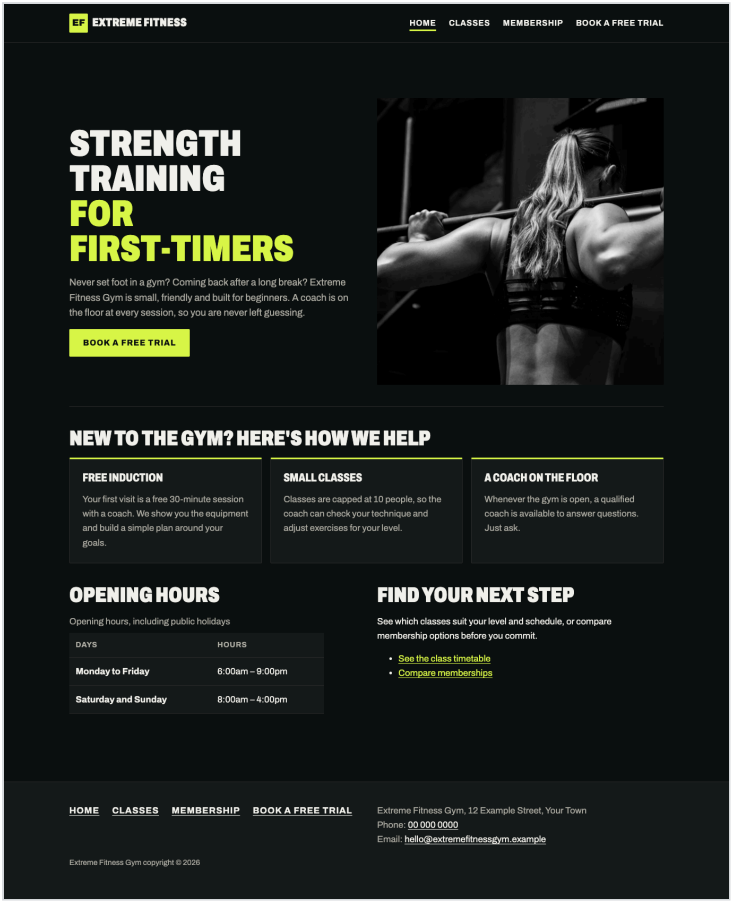


Figure 6. Home page, Chrome, 1366px.

The Home page tells visitors who the gym is for. A large heading sits next to the photo, with the trial button right underneath, so the main action is visible without scrolling.

Three cards explain the perks a nervous beginner cares about: a free induction, classes capped at 10 people and a coach always on the floor. An opening-hours table and links to the timetable and membership pages follow.

4b. Classes (classes.html)



Figure 7. Classes page, Chrome, 1366px.

The Classes page lists 15 weekly classes grouped by day. Each card gives the time, length, coach, level and a short description.

This page has my first JavaScript behaviour. The filter buttons (All classes, Beginner-friendly, Strength, Cardio, Mobility) hide classes that do not match, hide any day left empty, and update the "Showing X of 15 classes" message. It is the feature the site was built around: a beginner taps Beginner-friendly and the week shrinks to the 11 classes they can actually walk into.

4c. Membership (membership.html)



Figure 8. Membership page, Chrome, 1366px.

The Membership page compares three plans in cards: Casual ($15 per visit), Standard ($22 per week, marked "Most popular" with a lime border) and Student and Off-peak ($15 per week). A list shows what every plan includes.

The FAQ section is my second JavaScript behaviour. Clicking a question toggles its answer, keeping five answers to common worries on a short page. A lime banner at the bottom sends undecided visitors to the free trial.

4d. Book a free trial (join.html)

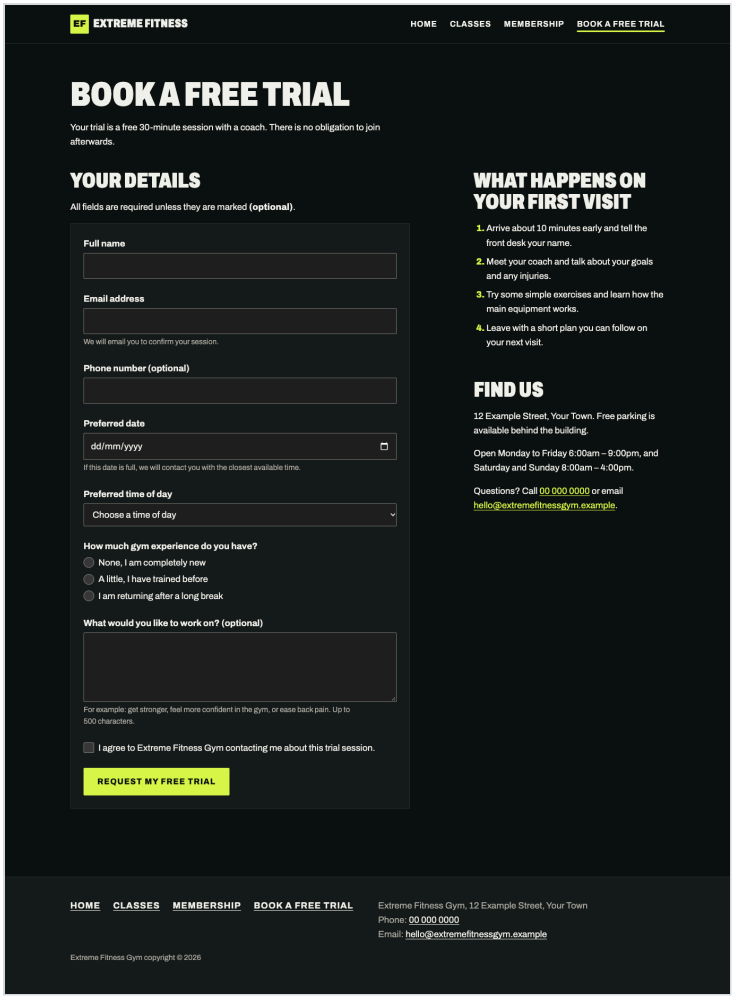


Figure 9. Book a free trial page, Chrome, 1366px.

The Join page explains the first visit in four numbered steps and holds the booking form. It asks for name, email, phone (optional), preferred date, time of day, experience level, goals (optional) and consent. Required fields use `required`, and inputs use matching types (`email`, `tel`, `date`).

The form uses `method="get"` and `action="thank-you.html"`, so every populated field is sent in the query string to the confirmation page (tested in Table 3).

5. WEB DESIGN BEST PRACTICES

I picked the three best practices that matter most for my audience: responsive layout, because most visitors are on phones; consistent navigation, because they are unsure where to look; and accessibility.

5a. Responsive page layout

MDN recommends building a simple single-column layout for narrow screens first, then adding a multi-column layout once there is enough width (MDN Web Docs, n.d.-b). I followed this. The base CSS is the phone layout, and the em-based breakpoints in Table 2 add columns. For the hero image I used `srcset` and `sizes` with 800px (38 KB) and 1600px (159 KB) WebP versions, so smaller screens do not download the large file (MDN Web Docs, n.d.-c). The image has `width` and `height` attributes to reserve its space. `object-fit: cover` with `object-position` keeps the lifter in frame whether the crop is 4:3 on mobile or square on desktop.

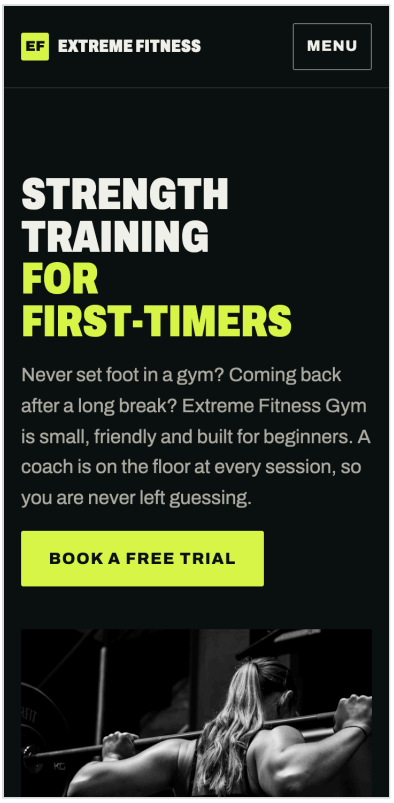


Figure 10. Home at 360px: single column, Menu button, heading and button first.

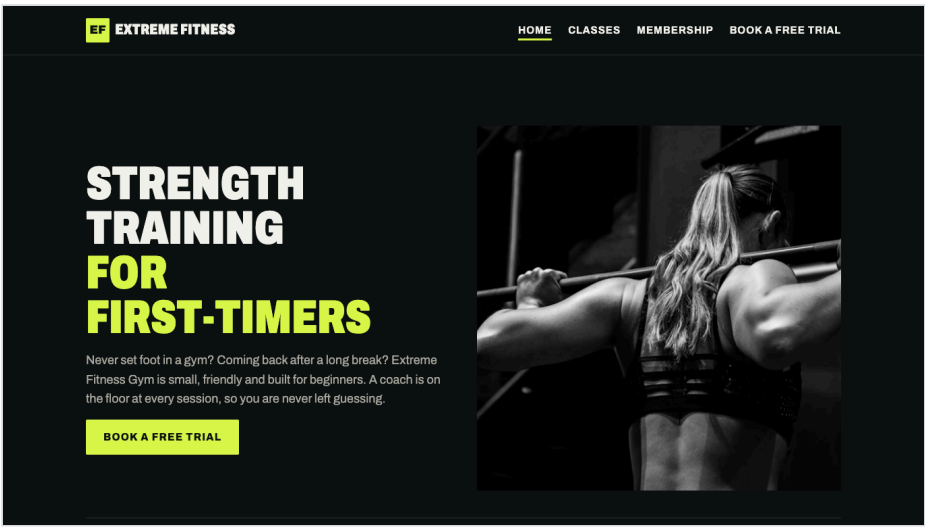


Figure 11. Home at 1366px: the same HTML laid out as two columns with CSS Grid.

5b. Navigation

Nielsen's "consistency and standards" heuristic says users should not have to wonder whether different words, situations or actions mean the same thing (Nielsen, 2024). I kept the header, menu order, labels and button styles identical on all five pages. The current page is marked with `aria-current="page"` and a thick lime underline, so the cue does not rely on colour alone. Below 52em the links collapse behind a Menu button that uses `aria-expanded`, closes with Escape and returns focus to the button. Filter buttons use `aria-pressed` and turn solid lime when active, and the FAQ arrow flips when an answer is open, so you can always tell what a click did.

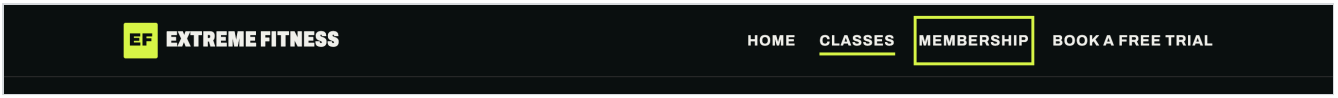


Figure 12. Classes page header: "Classes" is the current page (lime underline) and "Membership" has keyboard focus (3px lime outline).

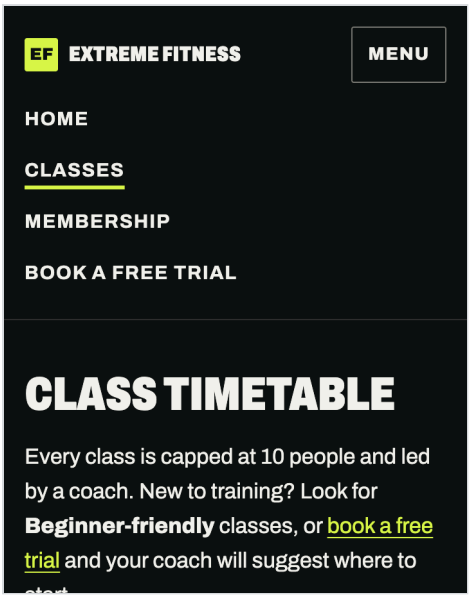


Figure 13. Menu open at 360px (`aria-expanded="true"`).

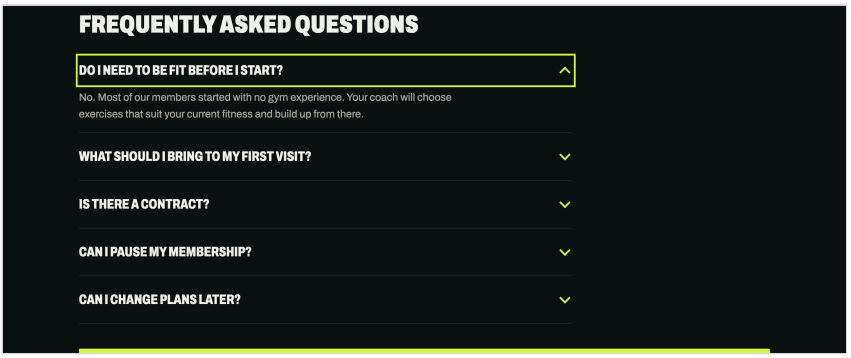


Figure 14. Membership FAQ: first question opened from the keyboard; the arrow flips and the button keeps its focus ring.

5c. Accessibility

WCAG 2.2 requires text alternatives, at least 4.5:1 contrast for normal text, reflow at 320 CSS pixels without scrolling in two directions, and a visible keyboard focus indicator (W3C, 2023). The hero photo has descriptive alt text; the decorative "EF" badge is hidden with `aria-hidden`. Every text pair is at least 7.6:1 (Table 1). The site reflows at 320px without sideways scrolling (6c). A skip link appears on the first Tab press, and every focusable element gets a 3px lime outline (dark inside the lime banner). The FAQ uses real `<button>` elements with `aria-expanded` and `aria-controls`. The filter result is announced through an `aria-live="polite"` region with `aria-atomic="true"`, so the whole sentence is read. If JavaScript fails, the full timetable, every FAQ answer and the full menu stay visible.

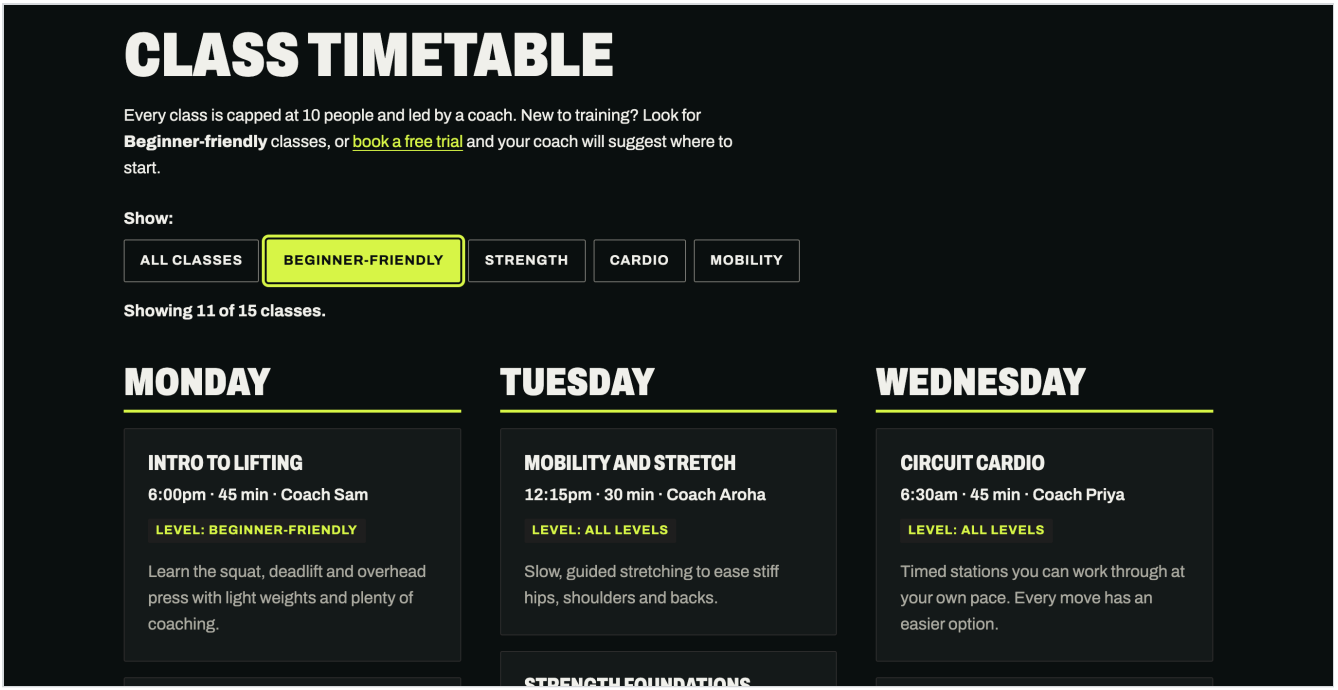


Figure 15. Beginner-friendly filter chosen with the keyboard: pressed state, focus ring and the live status "Showing 11 of 15 classes."

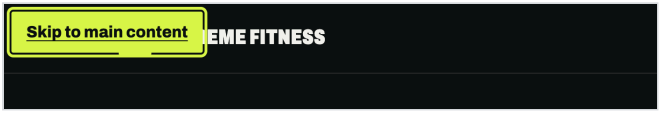


Figure 16. Skip link shown on the first Tab press.

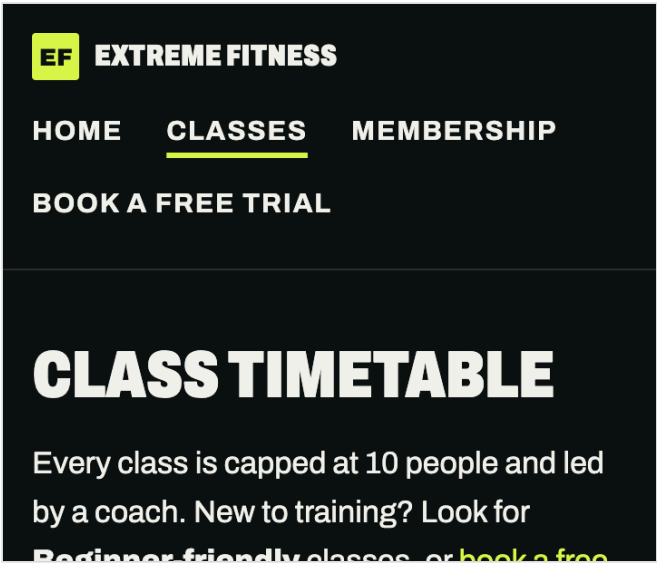


Figure 17. JavaScript disabled: full menu visible, filter hidden, all 15 classes listed.

6. TESTING

I wrote a test plan before I started clicking around and re-ran it after every fix. It covered all pages and links, the form, and the menu, filter and FAQ scripts (Table 3). I also compared each page's scroll width with the window width at 320, 360, 768, 1024 and 1366px: all 25 combinations had zero horizontal overflow.

Table 3. Functional test results (final version)

Test	Expected	Result
Header, footer and in-page links on all pages	Every link opens the right page; no missing files	**Pass.** All internal links resolve; no 404s in the console
Menu button (360px)	Opens the nav and sets `aria-expanded="true"`; Escape closes it and returns focus	**Pass.** Nav shown, then hidden; focus back on the Menu button
Class filter	Matching classes shown, empty days hidden, count and pressed state updated	**Pass.** Beginner 11, Strength 7, Cardio 4, Mobility 4, All 15 of 15
FAQ accordion	Answers start closed; a click opens one and sets `aria-expanded="true"`	**Pass.** 5 closed, then 1 open and 4 closed
Submit the empty form	Blocked, with a browser message on each required field	**Pass.** Name, email, date, time of day, experience and consent flagged; stays on join.html
Email "not-an-email"	Rejected by `type="email"`	**Pass.** "Please include an '@' in the email address."
Valid submission	Goes to thank-you.html with every field in the URL	**Pass.** `?name=Alex+Test&email=alex%40example.com&phone=&date=2026-11-12&session=evening&experience=none&goals=Get+stronger&consent=yes`
Past date (2020-01-01)	Ideally rejected	**Known limitation.** Accepted (see 6h and 8b)
JavaScript disabled	Content and navigation still usable	**Pass.** 15 classes and 5 FAQ answers visible; full menu shown; filter hidden

6a. Browser a: Google Chrome

Chrome 156 on macOS was my main test browser. All pages rendered as designed at every width and every behaviour worked. After the fixes in 6h there were no console errors. Section 4 screenshots are also from Chrome.

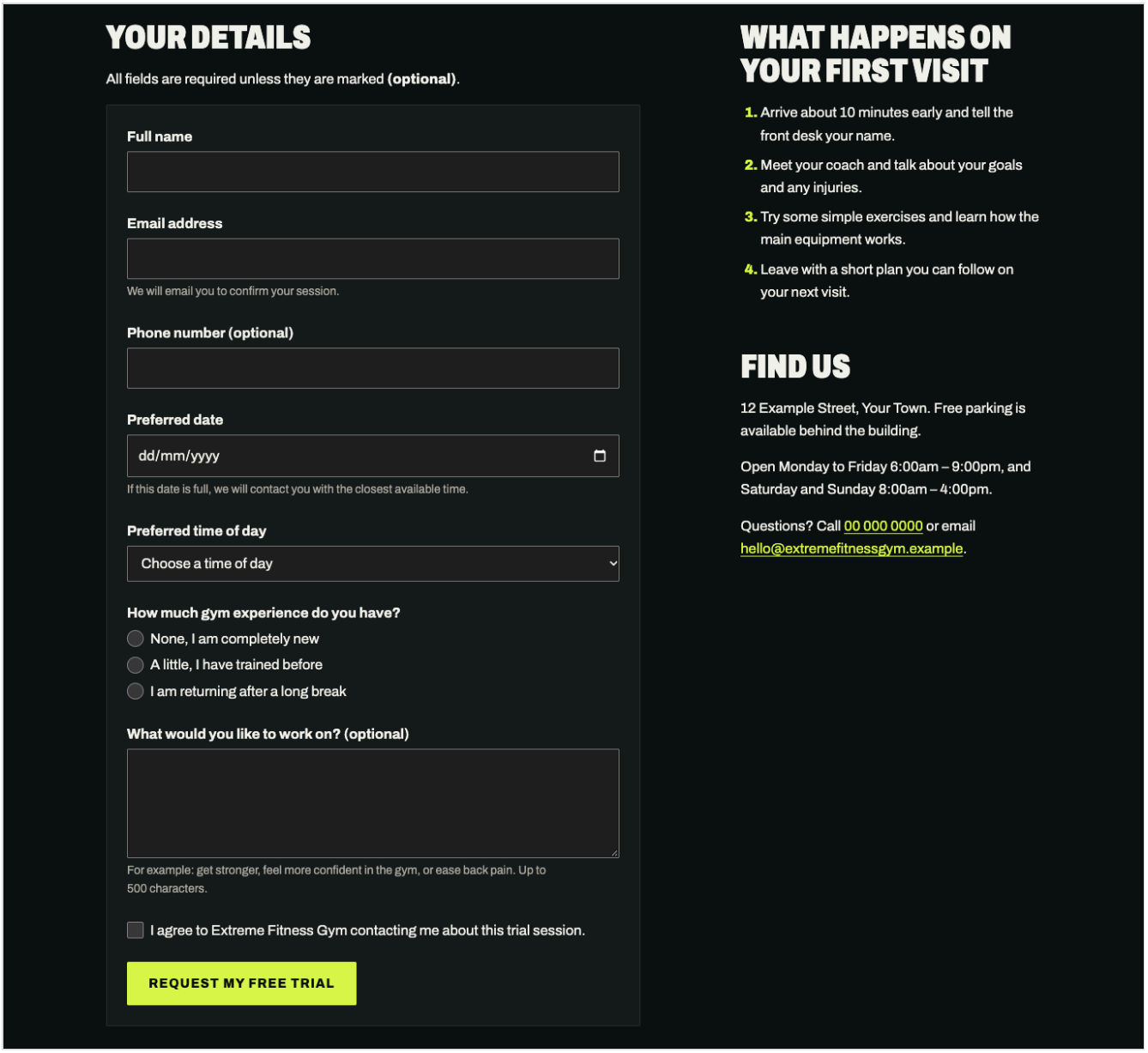


Figure 18. Chrome, 1366px: the Join page form beside the first-visit steps and "Find us".

6b. Browser b: Microsoft Edge

I repeated the same checks in Microsoft Edge. Because Edge and Chrome both use the Chromium engine, I did not see any visual differences in layout, font rendering or focus styles.

6c. Screen size a: 320px (small phone)

At 320px the header stays on one line with the Menu button, everything is a single column, and the filter buttons wrap onto extra rows. This width also stands in for 400% zoom on a 1280px window.

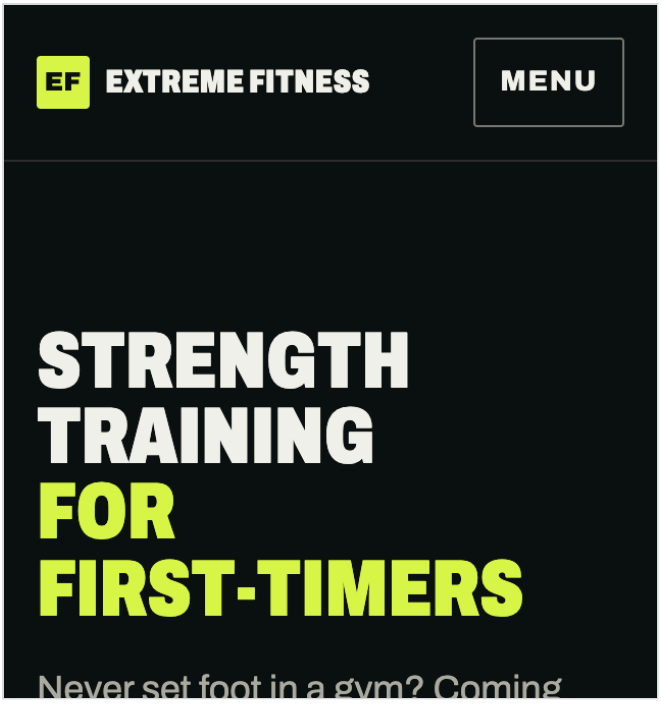


Figure 19. Home at 320px.

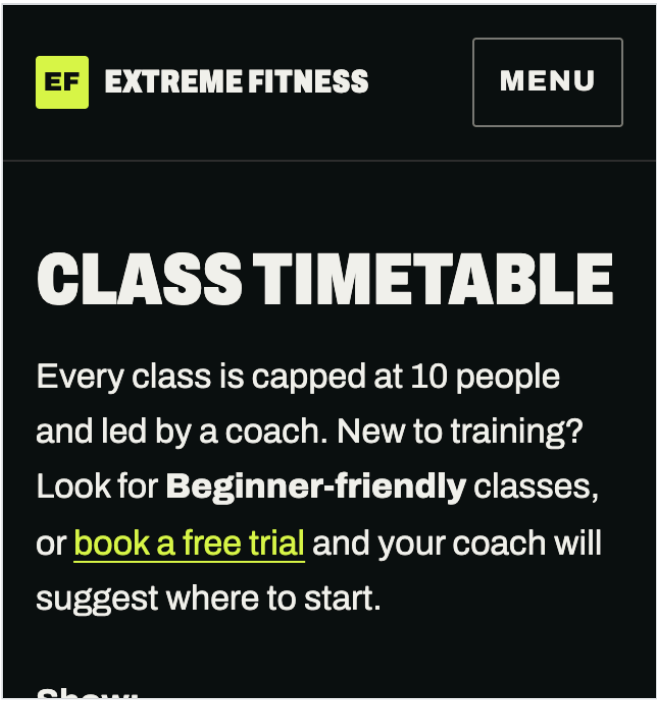


Figure 20. Classes at 320px with wrapped filter buttons.

6d. Screen size b: 768px (tablet portrait)

768px is below the 52em navigation breakpoint, so the Menu button is still used. This avoids the wrapped header I found earlier (6h). The timetable, "Every plan includes" list and footer switch to two columns, while the Home hero stays stacked until 56em.

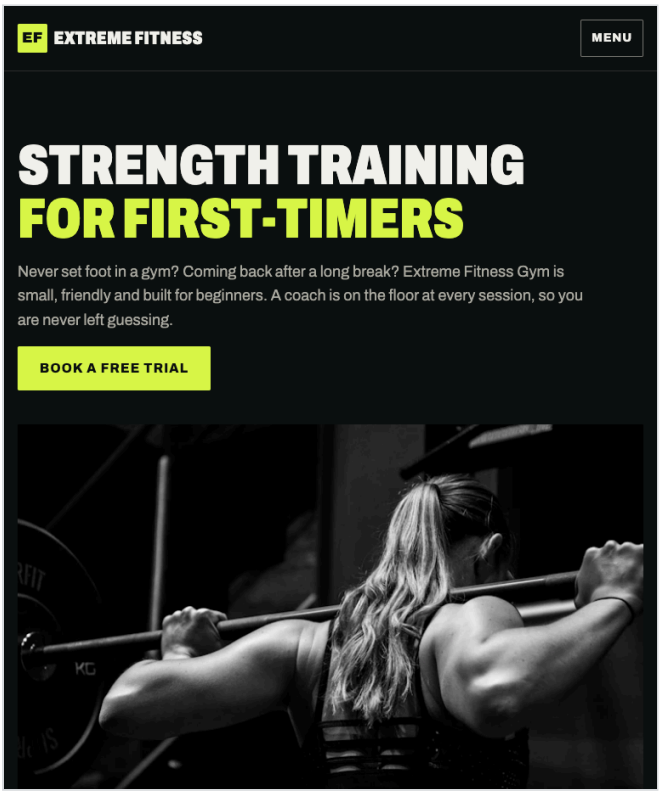


Figure 21. Home at 768px.

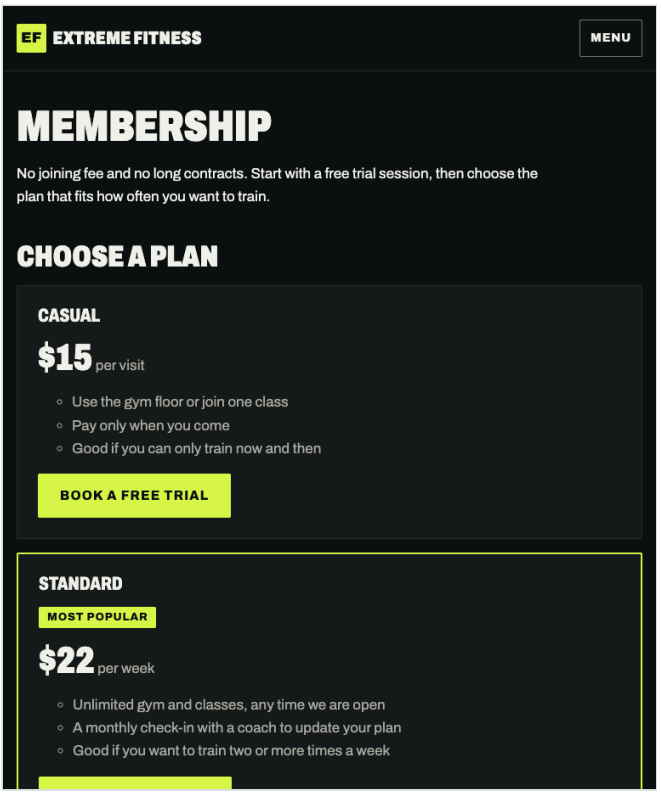


Figure 22. Membership at 768px.

6e. Screen size c: 1366px (laptop)

At 1366px the full menu spreads across the header, the hero text sits next to the photo, and the cards, plans and timetable form three columns. Content is capped at 70rem and centred.

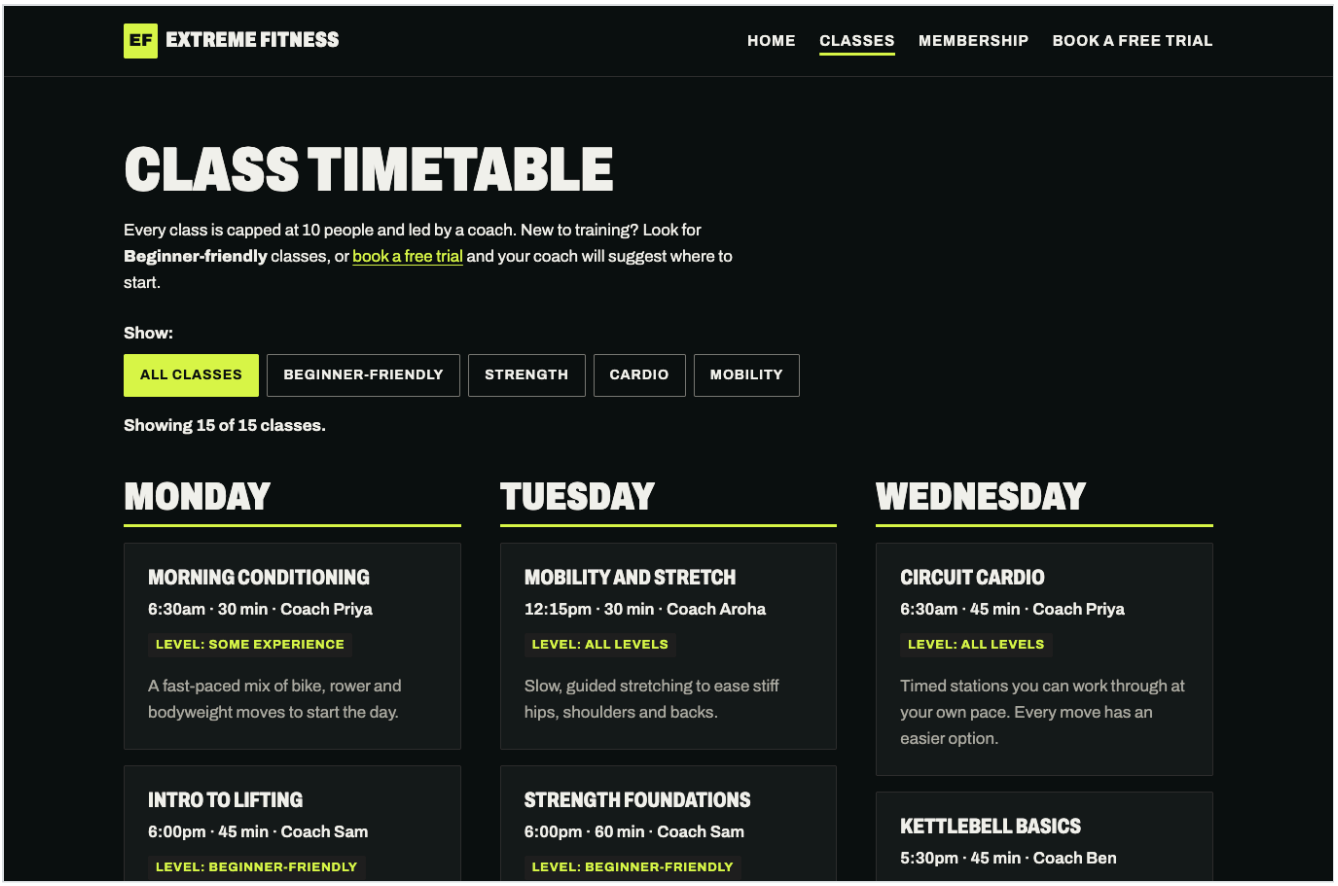


Figure 23. Classes at 1366px: three-column timetable.

6f. Mobile Lighthouse audit

I audited Home and Classes with Lighthouse 12.8.2 in Chrome using its default mobile settings (emulated phone with simulated throttling), once before and once after my refinements (Table 4).

Table 4. Mobile Lighthouse scores before and after refinement

Page	Run	Performance	Accessibility	Best Practices	SEO	CLS
Home	Before	100	100	96	100	0
	After	99	100	**100**	100	0
Classes	Before	100	100	96	100	0
	After	100	100	**100**	100	0

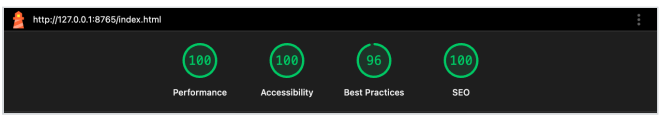


Figure 24. Before: Home, Best Practices 96.

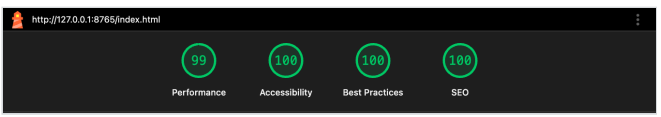


Figure 25. After: Home.

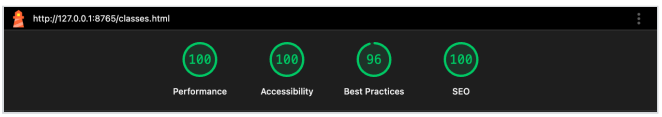


Figure 26. Before: Classes, Best Practices 96.

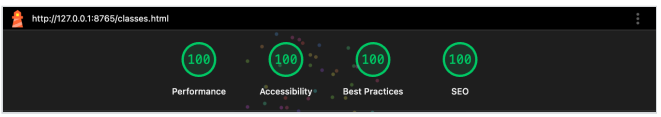


Figure 27. After: Classes.

Findings and refinements

1. Best Practices lost four points on both pages for console errors. Every page linked a favicon that did not exist (`favicon.ico` on four pages, `assets/images/favicon.svg` on Classes), and both returned 404 (figure below). I added `assets/images/favicon.svg`, linked it on every page, and Best Practices went to 100.
2. The web font had never loaded. This one was embarrassing to find so late. My Google Fonts link listed the weight axis twice (`Archivo:wght,wght@62..125,400..900`), Google answers that with HTTP 500, and every page had quietly fallen back to system fonts. I had been judging my condensed headings without ever seeing them. The API wants axes listed alphabetically by tag (Google, n.d.), so the link is now `Archivo:wdth,wght@…` and returns 200.
3. The filter bar shifted the layout, but only on a slow connection. Locally Lighthouse showed CLS 0 because `script.js` loads instantly, so I delayed the script by 1.5 seconds to fake a slow network. Then the problem showed up: JavaScript revealed the hidden filter bar and pushed the timetable down (CLS 0.208 at 360px, 0.054 at 1366px). I now reserve the bar's space in CSS. Under `@media (scripting: enabled)` (MDN Web Docs, n.d.-a) the hidden bar keeps its box with `visibility: hidden`, and the status line has a `min-height`, so the filter's shift is gone. At 360px a 0.081 shift remains when the header swaps the full menu for the Menu button. I left that, because the same code keeps the menu working without JavaScript.
4. Fixing the font broke something else. Once Archivo actually arrived, `display=swap` re-flowed the text, and Classes scored CLS 0.287 at 1366px on a throttled "Slow 4G" load. With `display=optional` the browser only uses the font if it is ready almost straight away, which rules out swap shifts (Hempenius & Pollard, 2022). On a slow first visit you see the system font and get Archivo from the cache on the next page. I can live with that. CLS went back to 0.
5. Lighthouse still flags render-blocking requests (the Google Fonts CSS and `styles.css`, estimated 710 to 970 ms). I kept the 13 KB stylesheet blocking because the page looks broken without it. Self-hosting the font with a preload is the fix I would do next. Compression is the host's job, and I left the CSS unminified so it stays readable.

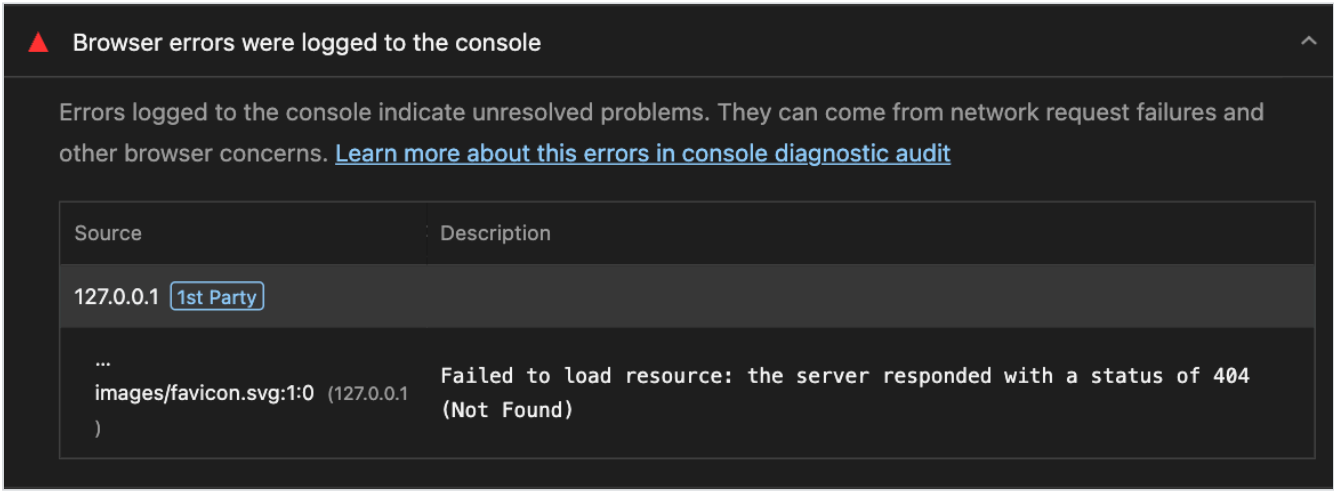


Figure 28. Lighthouse "before" run on Classes: the missing favicon logged as a console error.

6g. HTML/CSS validator

I uploaded all five HTML files to the W3C Nu HTML Checker (vnu 26.10.18). Each one returned "Document checking completed. No errors or warnings to show." `styles.css` passed the W3C CSS Validator (CSS level 3 + SVG) with no errors and one warning: "Due to their dynamic nature, CSS variables are currently not statically checked". This is expected because my colours and spacing are custom properties resolved at run time. I re-validated after every fix.

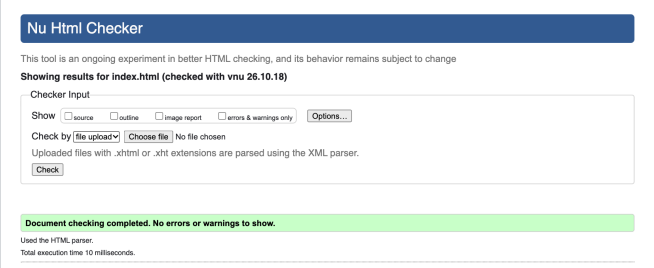


Figure 29. Nu HTML Checker: index.html.

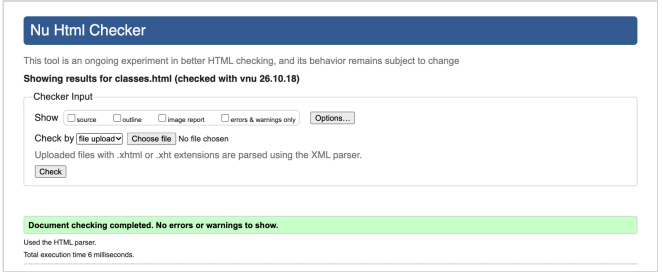


Figure 30. Nu HTML Checker: classes.html.

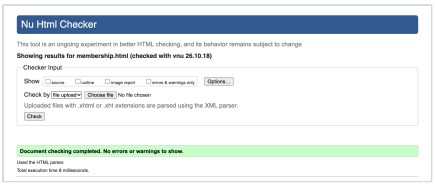


Figure 31. membership.html: no errors or warnings.

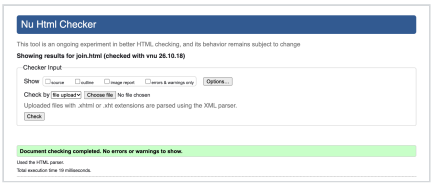


Figure 32. join.html: no errors or warnings.

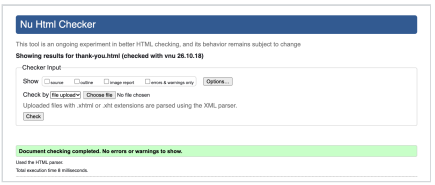


Figure 33. thank-you.html: no errors or warnings.

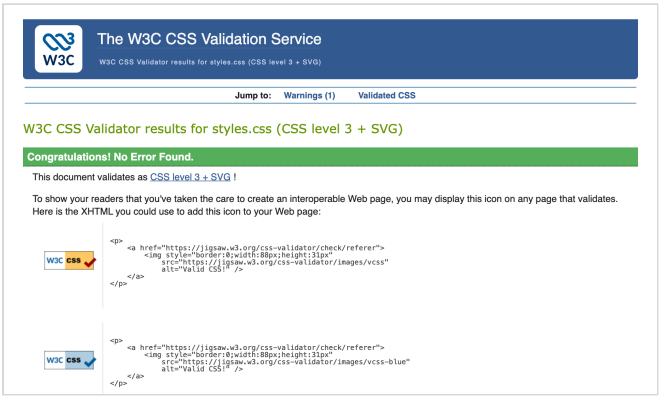


Figure 34. W3C CSS Validator: styles.css, no errors.

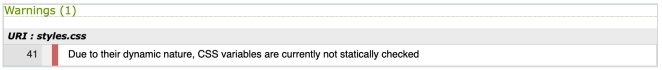


Figure 35. The single CSS warning (custom properties are not statically checked).

6h. Bugs found and fixed

Table 5 traces each issue from where I found it to the change and the retest.

Table 5. Issues found during testing, fixes and retest results

Issue	Fix	Retest
Menu button wrapped to a second line at 320px	Logo size set with `clamp()`; smaller logo and padding	**Pass:** one line at 320px
Full header wrapped at 768px	Navigation breakpoint moved from 48em to 52em	**Pass:** Menu button until 832px
Home heading broke awkwardly across lines	Heading sized with `clamp()` to fit its column; "first-timers" kept together with `white-space: nowrap`	**Pass** at all widths
Google Fonts URL `wght,wght` returned HTTP 500	Changed to `wdth,wght` on all five pages	**Pass:** 200; Archivo renders
Favicon 404 console errors	Added `assets/images/favicon.svg`; same link on every page	**Pass:** Best Practices 100
Filter bar pushed the timetable down	Space reserved in CSS; `min-height` on the status line	**Pass:** filter shift 0

Issue	Fix	Retest
Font swap shift once Archivo loaded	`display=swap` changed to `display=optional`	**Pass:** CLS 0 on Slow 4G
Page titles said "Fitness Extreme" while the logo said "Extreme Fitness"; generic corporate meta descriptions	Titles now "Page | Extreme Fitness Gym"; one description per page in the site's own tone	**Pass**
Filter status not always read as a full sentence	Added `aria-atomic="true"` to the live region	**Pass**
Date field accepts past dates	Not fixed. Needs a script that sets `min` to today's date	**Known limitation**

7. LAUNCH

Domain name

A WHOIS lookup on 29 October 2026 returned "Not found" for **extremefitness.co.nz**, so the name is available to register. MyHost, a New Zealand registrar, lists .co.nz registration at NZ$29.95 per year excluding GST (MyHost, n.d.-a). That is NZ$34.44 per year including 15% GST. A .co.nz name also tells local visitors the gym is based here.

Hosting options

I compared two free static hosts with a low-cost New Zealand shared host (Table 6).

Table 6. Hosting comparison (prices checked 29 October 2026)

	Netlify (Free plan)	GitHub Pages	MyHost Small (shared hosting)
Price	US$0; 300 credits per month, hard limit. Personal plan US$9/month for 1,000 credits	Free with a GitHub account	NZ$8.95/month or NZ$96.66/year + GST
Limits	15 credits per production deploy; 20 credits per GB of bandwidth	1 GB site; 100 GB/month soft bandwidth; 10 builds/hour	Unlimited storage; 2 websites
Custom domain + HTTPS	Yes, free SSL	Yes, free HTTPS	Yes, free automatic SSL
Forms	Netlify Forms; submissions are not charged credits	None (static files only)	Possible with a server-side script
Other features	Deploy from Git; custom headers via a `_headers` file; automatic compression	Deploy from a repository	Unlimited email accounts; email and ticket support
Business use	Allowed	"Not intended for or allowed to be used as a free web-hosting service to run your online business" (GitHub, n.d.)	Allowed

GitHub Pages is great for student portfolios, but its terms rule out business sites, so it is out. MyHost would work and includes email, but costs about NZ$111 a year including GST for features this site does not need. I would choose Netlify's Free plan (Netlify, n.d.-a). A first visit to Home is about 150 KB, so 300 monthly credits cover around ten deploys (150 credits) plus 7.5 GB of bandwidth (150 credits), enough for tens of thousands of page views (Netlify, n.d.-b). Netlify Forms could store real trial requests if the form used `method="post"` and a `data-netlify` attribute. The first-year cost would be just the domain: NZ$34.44.

8. MAINTENANCE

8a. Accessibility

I ran each check on all five pages of the final version (Table 7).

Table 7. Accessibility checks on the final website

Check	How I checked / finding	Result
Keyboard	Tabbed through every page. The skip link, header links, Menu button, filter buttons, FAQ buttons and every form control are reachable in a logical order and work with Enter/Space; Escape closes the menu and returns focus.	**Pass**
Visible focus	3px lime `:focus-visible` outline with a 2px offset on every control. Inside the lime CTA band the outline switches to dark so it does not disappear.	**Pass**
Headings and landmarks	One `h1` per page and no skipped levels (h1, h2, h3). `header`, `main` and `footer` on every page, with two `nav` elements labelled "Main" and "Footer".	**Pass**
Form labels and feedback	Every input has a `<label for>`; radio buttons are grouped in a `fieldset` / `legend`; hints linked with `aria-describedby`; "(optional)" in labels. Browser messages appear on the first invalid field. Finding: past dates are accepted.	**Pass, 1 known issue**
Status messages	Filter count is in an `aria-live="polite"` region. Finding: no `aria-atomic`, so a screen reader could read only the changed number. Added `aria-atomic="true"`.	**Fixed**
Image alternatives	Hero photo has descriptive alt text; the decorative "EF" badge is `aria-hidden`; there are no other content images.	**Pass**
Colour contrast	All text pairs measured (Table 1): lowest 7.62:1. Borders of controls 3.73:1. The current page uses an underline as well as colour.	**Pass**
Zoom and reflow	Tested at 320 CSS px (equal to 400% zoom on a 1280px window) and 683 to 768px (200% zoom). No horizontal scrolling, nothing cut off, header on one line.	**Pass**

8b. Feature improvements

Live booking

Right now the form just sends a GET request to a static page, so nothing is actually booked. A real gym needs a backend that caps each class at 10 people, shows how many spots are left on the timetable and emails a confirmation. The date picker could then offer only days the gym is open, which would also fix the past-date problem.

A "my week" planner

I think beginners struggle more with sticking to a routine than with any single class. Letting them star classes, save them with `localStorage` and export them to a calendar file (.ics) would help with that. Most of the groundwork is already there, because every class card has its tags, time and duration in the HTML.

9. CONCLUSION

Extreme Fitness does what I set out to do. A nervous beginner can find out whether this gym is for them, what it costs and what the first visit looks like, mostly from a phone. It is built from semantic HTML, one mobile-first stylesheet and three small JavaScript behaviours (menu, class filter and FAQ), and the content still works with JavaScript turned off. All pages pass the W3C HTML checker, the CSS has no errors, and both audited pages score 99 or 100 in every Lighthouse category.

Testing changed the site more than I expected. The early bugs were the obvious kind, like text wrapping where my breakpoints did not expect it. The later ones were not. My font link had been failing the whole time, and fixing it caused a layout shift that only appeared on a slow connection. I would not have found either by looking at my own screen. Two limitations remain: the form does not store anything, and the date picker still allows past dates. If I did this again, I would test on a throttled connection and a 320px window from the first week instead of the last.

10. REFERENCING

GitHub. (n.d.). *GitHub Pages limits*. GitHub Docs. Retrieved October 29, 2026, from https://docs.github.com/en/pages/getting-started-with-github-pages/github-pages-limits

Google. (n.d.). *CSS API update*. Google Fonts for Developers. Retrieved October 29, 2026, from https://developers.google.com/fonts/docs/css2

Hempenius, K., & Pollard, B. (2022, October 4). *Best practices for fonts*. web.dev. https://web.dev/articles/font-best-practices

MDN Web Docs. (n.d.-a). *scripting CSS media feature*. Mozilla. Retrieved October 29, 2026, from https://developer.mozilla.org/en-US/docs/Web/CSS/Reference/At-rules/@media/scripting

MDN Web Docs. (n.d.-b). *Responsive web design*. Mozilla. Retrieved October 29, 2026, from https://developer.mozilla.org/en-US/docs/Learn_web_development/Core/CSS_layout/Responsive_Design

MDN Web Docs. (n.d.-c). *Using responsive images in HTML*. Mozilla. Retrieved October 29, 2026, from https://developer.mozilla.org/en-US/docs/Web/HTML/Guides/Responsive_images

MyHost. (n.d.-a). *Domain names*. Retrieved October 29, 2026, from https://www.myhost.nz/domains/

MyHost. (n.d.-b). *Web hosting*. Retrieved October 29, 2026, from https://www.myhost.nz/hosting/web-hosting

Netlify. (n.d.-a). *Pricing and plans*. Retrieved October 29, 2026, from https://www.netlify.com/pricing/

Netlify. (n.d.-b). *Credit-based pricing plans*. Netlify Docs. Retrieved October 29, 2026, from https://docs.netlify.com/manage/accounts-and-billing/billing/billing-for-credit-based-plans/credit-based-pricing-plans/

Nielsen, J. (2024, January 30). *10 usability heuristics for user interface design*. Nielsen Norman Group. https://www.nngroup.com/articles/ten-usability-heuristics/

World Wide Web Consortium. (2023, October 5). *Web Content Accessibility Guidelines (WCAG) 2.2*. W3C Recommendation. https://www.w3.org/TR/WCAG22/

Assets

Archivo typeface: Omnibus-Type, via Google Fonts (https://fonts.google.com/specimen/Archivo), SIL Open Font License. Hero photograph: [add the photographer and source of hero-barbell-squat image, e.g. Unsplash/Pexels link] .

APPENDIX A: AI DECLARATION

The BIT507 brief allows limited AI use for planning, debugging, and generating placeholder text. I wrote all the HTML, CSS and JavaScript myself, and wrote the first draft of this QA document. I used an LLM for the following specific tasks (Tables 8 and 9):

Table 8. Generative AI use

Stage	What I asked for	What the AI produced	How I used or changed it	How I checked it
Placeholder content	Draft text for 15 gym classes, 3 membership tiers, and 5 FAQ answers	Realistic-sounding gym copy	I edited the tone to be less corporate and more welcoming, and formatted it into my HTML	Read through to ensure it made sense for a beginner
Debugging	Explaining why my `aria-live` region was not announcing filter updates	Explanation of `aria-atomic` and `aria-relevant` attributes	I added the correct attribute (`aria-atomic="true"`) to the filter status	Tested with a screen reader to confirm it spoke the updates
Planning	Brainstorming target audience pain points	A list of common gym anxieties	I used the ideas to write my conceptualisation section	Checked against my own gym experience

AI use in reviewing and preparing this document

This table continues the declaration above.

Table 9. Generative AI use (continued)

Stage	What I asked for	What the AI produced	How I used or changed it	How I checked it
QA review and document	Compare my QA draft with my final code and turn it into this PDF	Found fixes described in my draft that were missing from the code and applied them (font URL, favicon, filter-bar layout shift, `aria-atomic`, page titles, `display=optional`). Ran Lighthouse, the W3C validators and screen-size tests; took the Chrome screenshots; drew the site map and wireframes; edited and laid out this document	Reviewed every code change in `git diff`; kept my own wording wherever it matched the code	Re-ran Lighthouse and both validators on the final files; retested the menu, filter, FAQ and form

I can explain every line of code in my project and built the layouts and interactive features myself.